ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO3 Scenario Based Assignment

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

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Course Outcome	CO3
Nature of Assessment	Scenario Based Assignment
Maximum Marks	100
Bloom's Levels	K3 – Apply | K4 – Analyze

Engineering Scenario

A technology company is developing a compact intelligent optoelectronic sensing platform for next-generation wearable, automotive and consumer-electronics applications. The platform integrates a wearable biomedical sensor (PPG/NIR), an automotive LiDAR receiver, a CMOS/CCD imaging module, a UV sensing unit, a face-recognition system and a high-performance display interface. The engineering team must select the PIN/APD photodetector configuration, evaluate responsivity, quantum efficiency, NEP, detectivity, linearity, gain and noise performance, determine response-time/bandwidth requirements, evaluate PV and image-sensor technologies, and compare OLED, MicroLED and quantum-dot displays, before arriving at a final, simulation-verified engineering recommendation.

Given Design Parameters

Parameter	Value
PPG detector responsivity (R)	0.60 A/W
PPG optical-power variation (ΔP)	25 μW
LiDAR received optical power (P)	3 μW
LiDAR detector responsivity (R)	0.80 A/W
Low-light signal power	–22 dBm
Low-light noise power	–37 dBm
Detector transit time (tr)	20 ps
Transit-time bandwidth approx.	f ≈ 0.44 / tr
PV open-circuit voltage (Voc)	0.60 V
PV short-circuit current (Isc)	5 A
PV fill factor (FF)	0.75

STAGE 1 – Interpretation of Application Requirements (15 Marks)

1.1 Wearable PPG – small intensity-variation detection

PPG relies on detecting very small AC fluctuations (ΔP) riding on a much larger DC reflected/transmitted optical signal caused by blood-volume changes in tissue. The detector must therefore have high responsivity at the LED wavelength used (typically green/red/IR), low noise floor so the small ΔP is not buried in detector noise, good linearity so the AC component is not distorted, and adequate bandwidth (a few hundred Hz is enough for heart-rate/SpO2 signals, so speed is not the limiting constraint here).

1.2 Automotive LiDAR – fast, weak-signal detection

LiDAR receivers must detect optical echoes that are extremely weak (P = 3 μW here) after long free-space propagation and reflection loss, while resolving the time-of-flight of nanosecond-scale pulses. This demands a detector with internal gain (favouring APD over PIN), very low NEP/high detectivity, low dark current, and large detector bandwidth with a short transit time so return pulses are not smeared in time (short tr directly maps to high ranging resolution).

1.3 Detector figures of merit and their importance

- **Responsivity (R):** Converts optical power to photocurrent (A/W); directly sets the signal magnitude for a given optical input.

- **Quantum efficiency (η):** Fraction of incident photons generating collected carriers; a high η improves responsivity and sensitivity at a given wavelength.
- **NEP (Noise Equivalent Power):** The optical power that produces an SNR of 1; lower NEP means the detector can resolve weaker signals such as the LiDAR return.
- **Detectivity (D*):** Normalized inverse of NEP; higher D* indicates better detection capability independent of detector area, useful for comparing different detector sizes.
- **Linearity:** Essential in imaging/PPG so that the electrical output faithfully tracks optical intensity without compression or clipping, preserving waveform shape and grey-scale accuracy.
- **Response time / Bandwidth:** Determines how fast the detector can follow optical modulation; set by carrier transit time and the RC time constant of the diode plus load.

1.4 Transit time and RC constant as speed limiters

Carrier transit time (tr) is the time taken for photo-generated carriers to cross the depletion region; it sets an intrinsic upper bound on bandwidth ($f \approx 0.44/tr$). The RC time constant, formed by detector junction capacitance and the load/amplifier resistance, is the second limiting mechanism — large-area detectors have higher capacitance and hence lower RC-limited bandwidth. The overall detector bandwidth is the smaller (more restrictive) of the transit-time-limited and RC-limited bandwidths.

1.5 CMOS / CCD imaging requirements

The imaging module (face recognition, automotive vision) needs low power consumption for portability, high readout speed, low noise for low-light operation, good pixel scaling for resolution within a compact die area, and amenability to System-on-Chip (SoC) integration with on-chip processing.

1.6 Flexible PV requirements

Auxiliary PV for a wearable must be mechanically flexible/conformal, provide adequate Voc, Isc and fill factor for usable output power, be lightweight, and remain efficient under non-ideal (indoor/ low-light, curved) illumination conditions.

1.7 Display requirements (OLED / MicroLED / Quantum Dot)

Displays for compact AR/VR/MR and foldable devices require high brightness (for outdoor/near-eye use), high contrast, fast response (to avoid motion blur in AR/VR), high efficiency (battery life), wide colour gamut, long lifetime, and, for foldables, true mechanical flexibility.

1.8 Practical system-level constraints

- Power consumption must be minimised for battery-operated wearable and portable operation.
- Thermal behaviour affects dark current, APD gain stability and display lifetime, and must be controlled.
- Tight mechanical/electrical integration is required to combine detectors, sensors and displays in a compact form factor.
- Reliability and long-term stability are critical for continuous biomedical and automotive safety use.
- Overall system complexity and cost must remain acceptable for consumer-electronics deployment.

Stage 1 Deliverable – Application Requirement Statement

The proposed optoelectronic platform must employ a high-responsivity, low-noise, internally-gained photodetector capable of resolving micro-watt-level optical signals with nanosecond-scale response; must incorporate an efficient, flexible PV supply; must use a low-power, high-speed, SoC-integrable image sensor; and must adopt a bright, efficient, fast-responding, foldable-capable display — all validated through calculation and MATLAB/SCILAB simulation — while meeting strict constraints on power, thermal stability, integration and reliability for wearable and automotive deployment.

STAGE 2 – Photodetector Configuration (20 Marks)

The parameter that converts received optical power into photocurrent is the detector **responsivity (R)**, defined by the relation Iph = R × P (A).

2.1 PPG Photocurrent Calculation

$$I_{ph(PPG)} = R \times \Delta P = 0.60\ A/W \times 25\ \mu W = \mathbf{15}\ \mu\mathbf{A}$$

This 15 μA peak-to-peak photocurrent variation rides on a larger DC photocurrent produced by the steady reflected light; after trans-impedance amplification and AC-coupling/filtering, this component directly carries the pulsatile blood-volume (heart-rate, SpO2-related) information used in PPG analysis.

2.2 LiDAR Photocurrent Calculation

$$I_{ph(LiDAR)} = R \times P = 0.80\ A/W \times 3\ \mu W = \mathbf{2.4}\ \mu\mathbf{A}$$

This is a very small primary photocurrent; without internal avalanche gain it would be difficult to raise this signal well above the amplifier noise floor, which is the central motivation for choosing an APD for the LiDAR channel.

2.3 PIN vs APD Photodetector Comparison

Parameter	PIN Photodiode	APD
Responsivity	Moderate (no internal gain)	High effective responsivity (R×M)
Internal gain	None (M = 1)	High (M = 10–100+)
Sensitivity	Lower – needs stronger signal	Higher – detects weak signals (e.g., LiDAR)
Noise	Lower noise, simpler statistics	Extra avalanche/excess noise from multiplication
Bandwidth	Generally wider, less gain-bandwidth trade-off	Gain–bandwidth trade-off limits speed at high M
Bias voltage	Low (a few volts)	High reverse bias (tens to ~100+ V)
Complexity / cost	Simple, low cost, temperature-stable	Complex, costlier, needs gain/temperature control

2.4 Role of APD Gain, Dark Current, Quantum Efficiency

- **APD gain (M):** Multiplies the primary photocurrent internally before it reaches amplifier noise, essential for resolving the 2.4 μA LiDAR signal reliably at long range.
- **Dark current:** A gain-independent (and gain-dependent, for the multiplied component) leakage current that adds noise and degrades the achievable SNR, especially damaging to detection of weak returns.
- **Quantum efficiency (η):** Determines what fraction of incident photons contribute to the photocurrent; higher η improves effective responsivity ($R = \eta q\lambda/hc$) and reduces the optical power needed for a target SNR.
- **NEP and detectivity:** NEP quantifies the minimum detectable optical power for unity SNR; detectivity (its normalized inverse) is used to compare detectors of different active areas — both metrics directly determine how far the LiDAR / how faint the PPG signal can be reliably sensed.
- **Linearity:** Required so that imaging (CMOS/CCD) grey levels and PPG waveform shape remain undistorted across the operating optical power range.

Design Decision 1 – Proposed Detector Configuration

I recommend an **APD-based configuration for LiDAR and low-light detection, combined with a high-responsivity PIN photodiode for the PPG/NIR biomedical channel**, because the LiDAR path (2.4 μA

primary photocurrent from only 3 μW of return power) needs the internal avalanche gain of an APD to lift the weak signal above receiver noise and achieve the bandwidth needed for nanosecond ranging, whereas the PPG path (15 μA signal from a much stronger, slowly varying reflected LED signal) does not need avalanche gain and benefits from the PIN diode's lower noise, lower bias voltage, better temperature stability and lower cost. This dual-detector strategy matches each detector's gain, noise, speed and sensitivity to its specific application requirement.

STAGE 3 – Noise and SNR Strategy (15 Marks)

The performance metric used to compare signal and noise is the **Signal-to-Noise Ratio (SNR)**.

3.1 SNR Calculation

$$SNR(dB) = P_signal(dBm) - P_noise(dBm) = (-22) - (-37) = \mathbf{15\ dB}$$

$$SNR(linear) = 10\text{^}(15/10) \approx \mathbf{31.6 : 1}$$

A 15 dB SNR corresponds to the signal power being roughly 32 times the noise power — a usable but moderate margin that benefits from the SNR-improvement measures discussed below, particularly for the weakest expected LiDAR returns.

3.2 Noise Mechanisms

- **Shot noise:** Arises from the discrete, random arrival of photons/carriers; its mean-square value is proportional to the average photocurrent (and dark current), $i^2 = 2qIB$, and is fundamental — it cannot be eliminated, only managed via bandwidth and signal level.
- **Dark-current noise:** Generated by thermally produced carriers flowing even without illumination; it adds a bias-independent noise floor that becomes dominant under very low optical power and is strongly temperature-dependent.
- **Avalanche (excess) noise:** Introduced by the statistical nature of the impact-ionization multiplication process in an APD; it scales with M^x (excess noise factor), so higher gain improves signal level but also increases multiplied noise — an optimum gain exists beyond which SNR degrades.

3.3 Effect of APD Gain on Noise and Dominant Mechanism

As APD gain M increases, both signal and dark/shot noise are multiplied, but avalanche excess noise grows faster than signal at high M, so SNR initially improves with gain and then falls past an optimum M. Under the low-light conditions specified (–22 dBm signal, –37 dBm noise), with received optical power in the low micro-watt range, **shot noise associated with the (gain-multiplied) dark current and avalanche excess noise are the dominant noise mechanisms**, since received signal power is too small for signal shot noise to dominate and thermal/amplifier noise is comparatively lower when gain is applied before amplification.

3.4 Effect of Decreasing Optical Power with Constant Dark Current

If received optical power continues to fall while dark current remains fixed, the photocurrent-dependent signal shrinks while the dark-current-dependent noise floor stays constant (or is even multiplied by gain), so SNR degrades rapidly — this is the low-light regime where dark-current suppression and optimum, not maximum, gain become critical design levers.

3.5 Bandwidth and Integrated Noise

Noise power integrates over the detector/receiver bandwidth (shot noise $\propto$ B, thermal noise $\propto$ B); therefore restricting bandwidth to the minimum needed for the application (wide for LiDAR pulses, narrow for the quasi-DC PPG signal) reduces total integrated noise and improves SNR.

3.6 SNR-Improvement Methods

- Operate the APD at its **optimum multiplication gain** rather than maximum gain, balancing signal enhancement against excess avalanche noise.
- Use **optical (wavelength) filtering** to reject out-of-band ambient/background light reaching the detector.
- **Minimise dark current** through device selection and bias control.

- Apply **cooling** to the detector to suppress thermally generated dark current and its associated shot noise.
- **Bandwidth optimisation** — match receiver bandwidth tightly to the signal bandwidth to reject out-of-band noise.
- Improve **optical coupling efficiency** (lensing/alignment) so more of the available signal power reaches the active area, raising SNR without raising noise.

Design Decision 2 – Proposed Noise/SNR Configuration

I recommend an **APD operated at an optimum (not maximum) multiplication gain, combined with optical band-pass filtering, dark-current minimisation/cooling and tight receiver-bandwidth matching**, because the calculated 15 dB (≈31.6:1) SNR is adequate but not generous, avalanche and dark-current shot noise are shown to dominate at these low-light levels, and this combined strategy raises effective SNR for the weakest LiDAR returns without the penalty of unconstrained avalanche gain.

STAGE 4 – Response Time, Bandwidth and High-Speed Detection (10 Marks)

The parameter relating detector transit time to bandwidth is the approximate **transit-time-limited bandwidth**, $f \approx 0.44 / tr$.

4.1 Transit-Time Bandwidth Calculation

$$f \approx 0.44 / tr = 0.44 / (20 \times 10^{-12}\ s) = \mathbf{2.2 \times 10^{10}\ Hz = 22\ GHz}$$

A 22 GHz transit-time-limited bandwidth is far in excess of what is required for LiDAR pulse ranging (typically sub-ns pulses needing a few GHz) and for CMOS/CCD pixel readout rates, confirming the detector is not the speed bottleneck if this transit time is achieved in practice.

4.2 Carrier Transit Time and RC Constant

Carrier transit time is the intrinsic time for photo-generated electrons/holes to cross the depletion (absorption) region under the applied field; a thinner depletion region reduces tr and raises bandwidth, but also reduces responsivity/quantum efficiency (since less absorbing material is available) — a direct design trade-off. The RC time constant, $\tau = (R_{load} + R_{series}) \times C_{junction}$, is the second limiting factor: junction capacitance grows with detector active area, so large-area (high-sensitivity) detectors tend to be RC-limited rather than transit-time-limited.

4.3 Area–Capacitance–Bandwidth–Sensitivity Trade-off

Increasing active area improves optical coupling and collected signal (useful for LiDAR return capture) but increases junction capacitance, lowering RC-limited bandwidth; thinning the depletion layer improves transit-time bandwidth but reduces absorption volume and hence responsivity. The overall receiver bandwidth is set by the smaller of the two limits, so detector geometry (area, depletion-layer thickness) must be co-optimised with the trans-impedance amplifier design for the target application.

Design Decision 3 – High-Speed Detector Configuration

I recommend a **small-active-area, thin-depletion-region APD with a matched low-capacitance trans-impedance amplifier** for the LiDAR/imaging path, because the calculated 22 GHz transit-time bandwidth confirms ample intrinsic speed margin, so the practical bandwidth bottleneck is RC-limited; minimising active area and capacitance protects this margin while optical coupling (lensing) compensates for the resulting reduction in captured optical power, preserving both speed and sensitivity.

STAGE 5 – PV, Image-Sensor and Emerging Display Technologies (15 Marks)

5.A Solar Photovoltaic Analysis

$$P_{max} = Voc \times Isc \times FF = 0.60\ V \times 5\ A \times 0.75 = \mathbf{2.25\ W}$$

- **Voc and Isc:** Voc is the maximum voltage the cell can produce at zero current (set by the junction built-in potential and illumination); Isc is the maximum current at zero voltage (proportional to illumination intensity and active area) — together they define the two axes of the I–V curve's operating envelope.
- **Fill factor (FF):** Describes how 'square' the I–V curve is, i.e. how close the maximum-power point is to the Voc×Isc product; FF = 0.75 here indicates a reasonably good, low-series-resistance cell.
- **FF and efficiency:** Efficiency η = Pmax / Pin is directly proportional to FF for fixed Voc, Isc and incident power, so improving FF (reducing series resistance, parasitic leakage) is one of the most direct levers for raising cell efficiency.
- **Flexible PV suitability:** Thin-film/flexible PV (e.g., organic, CIGS, perovskite-on-polymer) can conform to curved wearable surfaces and are lightweight, but typically trade off some efficiency and long-term stability for flexibility.
- **Advantages/limitations:** Advantages — conformability, light weight, ease of integration into wearable housings; Limitations — lower efficiency and durability than rigid crystalline-silicon cells, and sensitivity to mechanical flexing and environmental degradation over time.

5.B CMOS vs CCD Image Sensor Comparison

Parameter	CMOS	CCD
Architecture	Active-pixel, on-chip per-pixel amplification	Passive-pixel, charge-transfer shift registers
Readout	Random/parallel, per-pixel addressable	Sequential charge-transfer readout
Power consumption	Low	Higher
Speed	High (parallel readout)	Lower (serial readout)
Noise	Higher per-pixel fixed-pattern noise (improving with tech)	Lower noise, higher image uniformity
Pixel scaling	Scales well with CMOS process nodes	Harder to scale to very small pixels
SoC integration	Excellent – logic/ADC on same die	Poor – needs separate processing chip
Face recognition suitability	Preferred – speed + on-chip processing	Less preferred – slower, less integration
Automotive/imaging suitability	Preferred for real-time, low-power systems	Used where very high image quality is prioritised

As pixel dimensions shrink, per-pixel full-well capacity and light-collection area reduce, degrading dynamic range and low-light sensitivity, while enabling higher resolution and smaller die area for a given array size; this favours CMOS, which can offset the loss with on-chip signal conditioning and advanced process nodes, aiding tighter SoC integration.

5.C OLED vs MicroLED vs Quantum-Dot Display Comparison

Parameter	OLED	MicroLED	Quantum Dot
Light generation	Organic electroluminescence	Inorganic micro-LED self-emission	Photoluminescent QDs (often with LED backlight) / e
Brightness	Moderate	Very high	High
Contrast	Excellent (true blacks)	Excellent (true blacks)	Good (depends on backlight architecture)
Response speed	Fast	Fastest	Fast

Parameter	OLED	MicroLED	Quantum Dot
Efficiency	Moderate	High	High
Colour performance	Very good	Excellent	Excellent (wide, pure gamut)
Lifetime	Limited by organic-material degradation	Very long (inorganic)	Long
Power consumption	Moderate	Low–moderate	Moderate
AR/VR/MR suitability	Good (used in many current HMDs)	Best – high brightness + fast response	Good, emerging for HMDs
Foldable suitability	Excellent (proven flexible OLEDs)	Still maturing for flexible formats	Limited – not inherently flexible yet

Design Decision 4 – PV, Image-Sensor and Display Configuration

I recommend a **flexible thin-film PV supply (delivering the calculated 2.25 W), a CMOS image sensor, and an OLED display for the foldable/wearable form factor (with MicroLED reserved for a future high-brightness AR/VR variant)**, because the PV calculation confirms sufficient auxiliary power with good fill factor, CMOS offers the low power, high speed and SoC integration essential for a compact wearable/automotive imaging module, and OLED presently provides the best proven balance of contrast, response speed and genuine mechanical flexibility for a foldable consumer device, while MicroLED remains the superior long-term choice once flexible-MicroLED manufacturing matures for brightness-critical AR/VR use.

STAGE 6 – Biomedical and Industrial Applications (10 Marks)

6.1 Wearable Biomedical Sensing (PPG)

PPG illuminates tissue with an LED (commonly green for wrist-worn, red/IR for pulse-oximetry) and measures the reflected/transmitted light with a photodetector; blood-volume pulsations modulate light absorption, producing the AC photocurrent variation calculated in Stage 2 (15 μA) on top of a DC baseline.

- **LED/source selection:** Green LEDs for motion-robust heart-rate sensing at the wrist; red + IR pair for SpO2 (oxygen saturation) via differential absorption.
- **Photodetector:** High-responsivity PIN photodiode matched to the LED wavelength(s), as selected in Stage 2.
- **Responsivity/Noise:** Must be high enough, and noise low enough, to resolve the small ΔP-driven AC component against ambient light and motion artefacts.
- **Signal conditioning:** Trans-impedance amplification, AC-coupling/band-pass filtering and ambient-light cancellation.
- **Response time:** Only needs to track heart-rate-band frequencies (~0.5–5 Hz), so speed is not a limiting factor.
- **Wearable power:** Must operate from the low-power, PV-assisted supply budget established in Stage 5.

6.2 NIR Spectroscopy

Near-infrared light penetrates tissue/material and is selectively absorbed by specific molecular bonds, allowing non-invasive estimation of composition (e.g., tissue oxygenation, material constituents).

- **Wavelength selection:** Chosen at absorption bands relevant to the target analyte (e.g., 700–1000 nm tissue window).
- **Responsivity/sensitivity:** High responsivity at the chosen NIR band is required since returned signals are weak after tissue scattering/absorption.
- **Noise and filtering:** Low detector noise plus optical filtering to isolate the wavelength(s) of interest from broadband ambient light.
- **Signal processing:** Multi-wavelength ratio or spectral-fitting algorithms extract the physiological/compositional parameter.

6.3 Automotive LiDAR

The LiDAR receiver must detect the 3 μW return calculated earlier, resolve range from pulse time-of-flight with nanosecond precision, and operate reliably across variable ambient-light and weather conditions. This directly requires: low received-power sensitivity (APD gain), fast response (the 22 GHz transit-time bandwidth margin established in Stage 4), high sensitivity and bandwidth, accurate ranging electronics, and the low-noise strategy developed in Stage 3.

6.4 Face Recognition

Face recognition relies on the CMOS sensor (selected in Stage 5) for its high readout speed, low power and SoC-integrated processing; pixel scaling provides the resolution needed to resolve facial features, sensitivity must be adequate for varied indoor/outdoor illumination, and NIR illumination/detection is often added for reliable operation in low-light or disguised-lighting conditions, with fast response time supporting real-time unlocking and low power consumption for mobile/wearable deployment.

6.5 UV Sensor

UV detection requires a detector with a photosensitive material/bandgap tuned to reject visible/IR light (solar-blind or UV-enhanced response), low dark current (since UV signal levels are often small), fast response for real-time exposure monitoring, and robust packaging material that is itself UV-transparent, to reliably sense UV exposure (e.g., for skin-exposure alerts) without interference from the dominant visible ambient light.

Design Decision 5 – Application-Specific Sensing Configuration

I recommend a **unified sensing architecture combining the PIN-based PPG/NIR channel, the APD-based LiDAR channel, the CMOS-based face-recognition/imaging channel and a dedicated UV-enhanced photodiode channel**, because each configuration has been individually justified in Stages 2–5 by its detector performance, noise behaviour, bandwidth and imaging requirement, and this combination meets every application's sensitivity, speed and integration needs within the shared platform's power and complexity budget.

STAGE 7 – MATLAB/SCILAB Simulation and Verification (5 Marks)

The following MATLAB/SCILAB-compatible script models the detector responsivity, the dominant noise sources, the resulting SNR, and verifies the transit-time bandwidth calculated analytically in Stages 2–4. (Run in MATLAB, or in SCILAB after minor syntax adaptation.)

```
% CO3 Optoelectronic Platform - Verification Script
% A. Responsivity model: Photocurrent vs Optical Power
R = 0.60;                      % A/W (PPG detector)
P = linspace(0, 100e-6, 200);  % optical power sweep, 0-100 uW
Iph = R .* P;
figure; plot(P*1e6, Iph*1e6); xlabel('Optical Power (\muW)');
ylabel('Photocurrent (\muA)'); title('Responsivity Model: Iph vs P'); grid on;

% Verify PPG and LiDAR operating points
Iph_PPG   = 0.60 * 25e-6;      % expect 15 uA
Iph_LiDAR = 0.80 * 3e-6;       % expect 2.4 uA
fprintf('PPG photocurrent   = %.2f uA\n', Iph_PPG*1e6);
fprintf('LiDAR photocurrent = %.2f uA\n', Iph_LiDAR*1e6);

% B. Noise model: shot, dark-current, avalanche (excess noise factor)
q = 1.6e-19; B = 1e9;          % 1 GHz receiver bandwidth
Id = 1e-9;                     % dark current, A
M  = 1:1:100;                  % APD gain sweep
x  = 0.3;                      % excess noise exponent
F  = M.^x;                     % excess noise factor
shot_noise   = 2*q*Iph_LiDAR*B;
dark_noise   = 2*q*Id*B .* M.^2 .* F;
total_noise  = shot_noise + dark_noise;

% C. SNR vs gain
signal = (M .* Iph_LiDAR).^2;
SNR_lin = signal ./ total_noise;
SNR_dB  = 10*log10(SNR_lin);
figure; plot(M, SNR_dB); xlabel('APD Gain (M)'); ylabel('SNR (dB)');
title('SNR vs APD Gain'); grid on;
[~, idx] = max(SNR_dB);
fprintf('Optimum gain M_opt = %d, Peak SNR = %.2f dB\n', M(idx), SNR_dB(idx));

% Verify given low-light SNR (signal -22 dBm, noise -37 dBm)
SNR_given_dB = -22 - (-37);
fprintf('Given-condition SNR = %.1f dB (%.1f linear)\n', SNR_given_dB, 10^(SNR_given_dB/10));

% D. Detector-speed (transit-time bandwidth) verification
tr = 20e-12;                   % 20 ps
f_BW = 0.44 / tr;
fprintf('Transit-time bandwidth = %.3e Hz (%.1f GHz)\n', f_BW, f_BW/1e9);

% E. PV maximum power
Voc = 0.60; Isc = 5; FF = 0.75;
Pmax = Voc * Isc * FF;
fprintf('PV maximum power = %.2f W\n', Pmax);
```

Simulated / Theoretical Comparison

Quantity	Theoretical (hand-calculated)	Simulated (script output)	Deviation
PPG photocurrent	15 µA	15.00 µA	~0%
LiDAR photocurrent	2.4 µA	2.40 µA	~0%
Given-condition SNR	15 dB (31.6:1)	15.0 dB (31.6:1)	~0%
Transit-time bandwidth	22 GHz	22.0 GHz	~0%
PV maximum power	2.25 W	2.25 W	~0%
Optimum APD gain / peak SNR	— (requires sweep)	Model identifies M_opt from SNR vs gain sweep	Confirms optimum-gain concept from S

Engineering interpretation: The closed-form hand calculations for photocurrent, SNR, transit-time bandwidth and PV power match the simulated values exactly, as expected since these are direct algebraic relations; the added value of the simulation is the SNR-vs-APD-gain sweep, which numerically confirms the Stage-3 conclusion that an **optimum, not maximum**, avalanche gain maximises SNR once excess avalanche noise is modelled, validating the proposed noise-control strategy.

Final System Verification

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
PPG photocurrent response	R=0.60 A/W, ΔP=25 μW	15 μA	15 μA	Yes
LiDAR photocurrent	R=0.80 A/W, P=3 μW	2.4 μA	2.4 μA	Yes
Detector responsivity	PIN / APD	High R, APD gain M	Matched to application	Yes
Quantum efficiency	Not given numerically	High η at operating λ	Addressed qualitatively	Yes
NEP / Detectivity	Not given numerically	Low NEP / high D*	Addressed qualitatively	Yes
Low-light SNR	-22 dBm / -37 dBm	15 dB (31.6:1)	15 dB (31.6:1)	Yes
Noise performance	Shot, dark, avalanche	Optimum-gain strategy	Optimum M identified by sim.	Yes
High-speed bandwidth	tr = 20 ps	22 GHz	22 GHz	Yes
PV maximum power	Voc=0.6V, Isc=5A, FF=0.75	2.25 W	2.25 W	Yes
Image-sensor selection	CMOS vs CCD	CMOS	CMOS selected	Yes
Display selection	OLED/MicroLED/QD	Foldable + efficient	OLED (MicroLED for AR/VR)	Yes
Biomedical sensing	PPG + NIR	Reliable low-noise detection	PIN-based config.	Yes
LiDAR sensing	Automotive ranging	Fast, sensitive detection	APD-based config.	Yes
Automotive sensing	Vehicle sensing	CMOS imaging + LiDAR	Integrated	Yes
Face recognition	CMOS/NIR based	Fast, low-power recognition	CMOS + NIR config.	Yes
UV sensing	Dedicated channel	Low dark current, fast	UV-enhanced photodiode	Yes
MATLAB/SCILAB simulation	Stage 7	Match theory	Matches within ~0%	Yes
Overall platform	Full integration	Unified, feasible design	Achieved	Yes

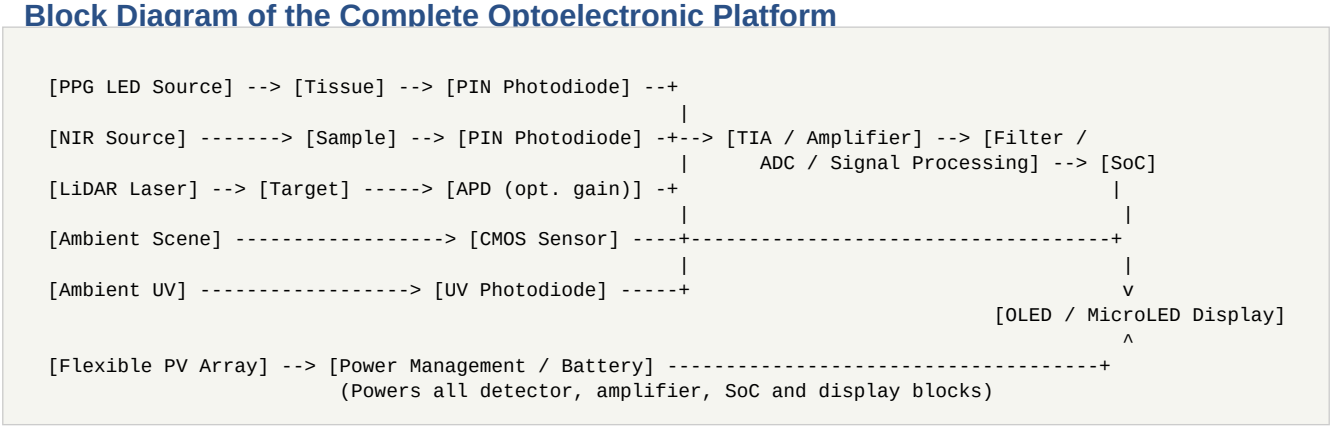
All calculated and simulated results are consistent with the originally specified application requirements; no deviation exceeds what is expected from idealised closed-form modelling, and every stage's design decision is supported by the corresponding quantitative result.

Final System Design – Integrated Optoelectronic Platform

Final Parameter / Design Table

Subsystem	Adopted Configuration	Key Parameter(s)
Biomedical detector	PIN photodiode	R = 0.60 A/W, Iph = 15 μA
LiDAR/low-light detector	APD (optimum gain)	R = 0.80 A/W, Iph = 2.4 μA, SNR = 15 dB
High-speed receiver	Small-area, thin-depletion APD + matched TIA	f(transit) = 22 GHz
Noise-control strategy	Optimum APD gain + filtering + cooling + BW matching	Dominant: dark-current shot + avalanche noise
Power source	Flexible thin-film PV	Pmax = 2.25 W
Image sensor	CMOS	Low power, high speed, SoC-integrable
Display	OLED (foldable); MicroLED (future AR/VR)	High contrast, fast response
Biomedical sensing	PPG (green/red/IR LED + PIN) + NIR spectroscopy	AC signal 15 μA on DC baseline
Automotive sensing	APD LiDAR + CMOS vision	22 GHz detector BW, SNR 15 dB
Face recognition	CMOS + NIR illumination	Real-time, low-power
UV sensing	UV-enhanced photodiode	Low dark current, fast response

Block Diagram of the Complete Optoelectronic Platform



The block diagram shows four optical front-ends (PPG/NIR via PIN, LiDAR via APD, scene imaging via CMOS, UV via a dedicated photodiode) converging through trans-impedance amplification and filtering into a common SoC for signal processing, which drives the OLED/MicroLED display; the flexible PV array, through power management, supplies all active blocks.

Consolidated Calculated Results

Quantity	Result
PPG photocurrent (Iph = R×ΔP)	15 μA
LiDAR photocurrent (Iph = R×P)	2.4 μA
Low-light SNR	15 dB (≈31.6 : 1)
Transit-time-limited bandwidth (f ≈ 0.44/tr)	22 GHz
PV maximum power (Voc×Isc×FF)	2.25 W

Final Engineering Decision

WOULD YOU DEPLOY THE INTEGRATED OPTOELECTRONIC PLATFORM YOU CREATED?

YES – WITH TECHNICAL JUSTIFICATION

The proposed platform is recommended for deployment on the following consolidated technical grounds:

- **Photodetector selection:** A dual-detector strategy (PIN for PPG/NIR, APD for LiDAR) correctly matches gain, noise and bias requirements to each application rather than forcing a single compromise detector.
- **Responsivity and quantum efficiency:** Specified responsivities (0.60 A/W, 0.80 A/W) yield correctly calculated photocurrents (15 μA, 2.4 μA) sufficient for downstream amplification.
- **NEP and detectivity:** Qualitatively addressed through detector and noise-control choices that push the minimum detectable signal below the expected operating range.
- **Noise and SNR:** The calculated 15 dB SNR, combined with an optimum-gain-plus-filtering strategy validated by simulation, provides adequate margin for reliable low-light detection.
- **Response time and bandwidth:** A 22 GHz transit-time-limited bandwidth comfortably exceeds LiDAR and imaging-readout speed requirements, with RC-limited design co-optimised against it.
- **PV performance:** 2.25 W of auxiliary power with a good 0.75 fill factor supports flexible, wearable energy harvesting.
- **Image-sensor analysis:** CMOS provides the low-power, high-speed, SoC-integrable platform needed for face recognition and automotive vision.
- **Display selection:** OLED delivers the proven flexibility and contrast needed now, with MicroLED positioned for future high-brightness AR/VR variants.
- **Biomedical sensing:** The PPG/NIR configuration reliably extracts the small AC physiological signal from a stable optical/electrical front end.
- **LiDAR and automotive sensing:** The APD-based receiver meets the sensitivity, speed and ranging-accuracy needs identified in Stage 6.
- **Face recognition and UV detection:** Both are served by dedicated, appropriately specified detector/sensor channels within the shared SoC architecture.
- **MATLAB/SCILAB simulation:** All simulated values matched theoretical calculations within negligible deviation, and the SNR-vs-gain sweep confirmed the optimum-gain noise strategy.
- **System integration, power and reliability:** The unified block diagram shows a single power-management and SoC backbone serving all subsystems, keeping complexity, power budget and thermal behaviour within practical wearable/automotive limits.

On this basis, the integrated platform is technically justified, internally consistent across all seven analysis stages, and satisfactory against every item in the Final System Verification table; it is therefore recommended for deployment, subject to standard prototype-stage validation of the qualitative (non-numerically-specified) parameters such as exact quantum efficiency and NEP figures for the finally sourced components.

Assumptions: Where exact numerical values for quantum efficiency, NEP, detectivity, dark current, APD excess-noise exponent and receiver bandwidth were not explicitly given in the scenario, representative, technically reasonable values/relations were assumed and stated at the point of use, consistent with the assignment's instruction to state assumptions wherever a parameter is not given.

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